

Thus, the full Lagrangian for the witness particle (relativistic charged particle in electromagnetic potentials ϕ and A) is:

$$L = -mc^2 \sqrt{1 - v^2/c^2} + q \vec{A} \cdot \vec{v} - q\phi, \quad (I)$$

such that:

$$S = \int L dt = \underbrace{-mc^2 \int \frac{dt}{\gamma}}_{S_{rel}} + \underbrace{q \int (\vec{A} \cdot \vec{v} - \phi) dt}_{S_{int.}}$$

$\bullet dS_{rel} = -mc^2 d\tau$
 $\sim$ a particle with mass accumulates action proportional to its proper time $d\tau$
 $\bullet dS_{int.} = q(\vec{A} \cdot d\vec{r} - \phi dt)$
 $\sim$ a charged particle accumulates action by moving through electromagnetic potentials

where $\vec{A}$ and ϕ are the electromag. potentials defined by the ME for the source particle together with the definitions $\vec{E} = -\vec{\nabla}\phi - \frac{\partial \vec{A}}{\partial t}$ and $\vec{B} = \vec{\nabla} \times \vec{A}$.
 which gives $\vec{E}$ and $\vec{B}$

Then, applying the Euler-Lagrange equations to (I):

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \vec{v}} \right) - \vec{\nabla} L = \frac{d}{dt} \left(-mc^2 \frac{-\vec{v}/c^2}{\sqrt{1-v^2/c^2}} + q\vec{A} \right) - q\vec{\nabla} (\vec{A} \cdot \vec{v} - \phi) = \frac{d}{dt} (\gamma m \vec{v} + q\vec{A}) - q\vec{\nabla} (A_z v_z - \phi)$$

$v_x, v_y \ll v_z$

$$\Rightarrow \frac{d}{dt} (\vec{p}_{rel} + q\vec{A}) = q\vec{\nabla} (A_z v_z - \phi)$$

Integrating both sides:

$$\left[\vec{p} + q\vec{A} \right]_{t=-\infty}^{t=+\infty} = \underbrace{\Delta \vec{p}}_{\substack{\text{momentum} \\ \text{change}}} = q \cdot \int_{-\infty}^{+\infty} \vec{\nabla} (A_z v_z - \phi) dt = q \vec{\nabla} \int_{-\infty}^{+\infty} (A_z v_z - \phi) dt$$

assumption that x, y are fixed with t
 gradient of a scalar function
 as we derived!
 $\left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right)$

Remembering that $\vec{W} = \frac{c}{qQ} (\Delta p_x, \Delta p_y, -\Delta p_z)$ and seeing that we can write $\Delta \vec{p} = \frac{qQ}{c} \vec{\nabla}_R W$, where

$$\vec{R} = (x_w, y_w, -z) \Rightarrow \Delta \vec{p} = \frac{qQ}{c} \left(\frac{\partial W}{\partial x_w}, \frac{\partial W}{\partial y_w}, -\frac{\partial W}{\partial z} \right), \text{ we have:}$$

$$\Rightarrow (w_x, w_y, w_z) = \frac{c}{qQ} \cdot \frac{qQ}{c} \left(\frac{\partial W}{\partial x_w}, \frac{\partial W}{\partial y_w}, -\frac{\partial W}{\partial z} \right) \Rightarrow \begin{cases} w_x = \frac{\partial W}{\partial x_w} \\ w_y = \frac{\partial W}{\partial y_w} \\ w_z = -\frac{\partial W}{\partial z} \end{cases} !$$

So, in short:

$$\Delta \vec{p} = \vec{\nabla} \left(\underbrace{\frac{q}{c} \int_{-\infty}^{\infty} |A_s v_s - \phi| dt}_{f(\vec{r})} \right) = \frac{q \cdot Q}{c} (w_x, w_y, -w_z)$$

$\text{kg m/s} \quad \text{kg m}^2/\text{s}^2 \cdot 1/c^2 = J/c^2 = V/c$

$$\Rightarrow (w_x, w_y, -w_z) = \frac{c}{qQ} \cdot \vec{\nabla} f = \vec{\nabla} \left(\frac{c}{qQ} \int_{-\infty}^{\infty} |A_s v_s - \phi| dt \right) = \vec{\nabla}_R W(\vec{\rho}_s, \vec{\rho}_w, z)$$

$\left(\frac{\partial}{\partial x_w}, \frac{\partial}{\partial y_w}, \frac{\partial}{\partial z} \right) = \left(\frac{\partial}{\partial x_w}, \frac{\partial}{\partial y_w}, \frac{\partial}{\partial(z-z)} \right)$
 $\sim z$

And thus:

$$(w_x, w_y, -w_z) = \vec{\nabla}_R W \Rightarrow \begin{cases} (w_x, w_y) = \vec{w}_\perp = \vec{\nabla}_{w,\perp} W \\ w_z = -\frac{\partial}{\partial(z)} W = \frac{\partial}{\partial z} W \end{cases}$$

Furthermore, this potential formalism enables directly obtaining the so-called Panofsky-Wenzel theorem from the second cross derivatives of W :

$$\frac{\partial \vec{w}_\perp}{\partial z} = \frac{\partial}{\partial z} (\vec{\nabla}_{w,\perp} W) = \vec{\nabla}_{w,\perp} \left(\frac{\partial}{\partial z} W \right) = \vec{\nabla}_{w,\perp} w_z \rightarrow \frac{\partial \vec{w}_\perp}{\partial z} = \vec{\nabla}_{w,\perp} w_z \quad (II)$$

↳ This property of the wakes is very useful as it enables calculating $\vec{w}_\perp$ from w_z by

a simple integration of (II):

$$\vec{w}_\perp = \int_{-\infty}^z \vec{\nabla}_{w,\perp} w_z dz^* \quad \text{Furthermore, } w_z = -\frac{c}{qQ} \Delta p_s = -\frac{c}{qQ} \int_{-\infty}^{\infty} \underbrace{(F_L)}_{\text{impulse approx.}} dt = -\frac{c}{Q} \int_{-\infty}^{\infty} dt E_s|_{s=ct-z}$$

$\vec{F}_L = q[E_s \hat{s} + (E_x - vB_y)\hat{x} + (E_y + vB_x)\hat{y}]$

⇒ All wake functions can be calculated from E_s !

$$w_z(\vec{\rho}_s, \vec{\rho}_w, z) = -\frac{c}{Q} \int_{-\infty}^{\infty} dt E_s(\vec{\rho}_s, \vec{\rho}_w, z, t) \Big|_{s=ct-z}; \quad \vec{w}_\perp = \int_{-\infty}^z \vec{\nabla}_{w,\perp} w_z dz^* !$$

Another important property of the wake potential W is being an harmonic function in the transverse coordinates x and y [if $\beta = 1 \sim v = \beta c$]*:

$$\nabla_{\perp}^2 W = \frac{\partial^2 W}{\partial x^2} + \frac{\partial^2 W}{\partial y^2} = 0 \quad \sim \quad \frac{\partial W_x}{\partial x} = -\frac{\partial W_y}{\partial y}$$

To prove this, we first remember that, from the ME in free space, the potentials ϕ and $\vec{A}$ satisfy the wave equation:

$$\nabla^2 \vec{A} - \overset{1/c^2}{\epsilon_0} \frac{\partial^2 \vec{A}}{\partial t^2} = 0; \quad \nabla^2 \phi - \epsilon_0 \frac{\partial^2 \phi}{\partial t^2} = 0$$

Now, since $W = q \int_{-\infty}^{\infty} (A_s v - \phi) dt$, we have:

$$\nabla^2 W = \nabla_{\perp}^2 W + \frac{\partial^2 W}{\partial z^2} = \nabla^2 \left(q \int_{-\infty}^{\infty} (A_s v - \phi) dt \right) = q \int_{-\infty}^{\infty} (c \nabla^2 A_s - \nabla^2 \phi) dt = q \int_{-\infty}^{\infty} \frac{1}{c^2} \frac{\partial^2}{\partial t^2} (v A_s - \phi) dt$$

$$\Rightarrow \nabla_{\perp}^2 W = q \int_{-\infty}^{\infty} \left(\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial z^2} \right) (v A_s - \phi) dt$$

$$\text{Furthermore, } \left(\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial z^2} \right) = \left(\frac{1}{c} \frac{\partial}{\partial t} + \frac{\partial}{\partial z} \right) \cdot \left(\frac{1}{c} \frac{\partial}{\partial t} - \frac{\partial}{\partial z} \right), \text{ and } z = \beta c t - z \Rightarrow \frac{d}{dt} = \left(\frac{\partial}{\partial t} + \frac{\partial z}{\partial t} \frac{\partial}{\partial z} \right) = \beta c \left(\frac{1}{\beta c} \frac{\partial}{\partial t} + \frac{\partial}{\partial z} \right),$$

meaning that, for $\beta = 1$:

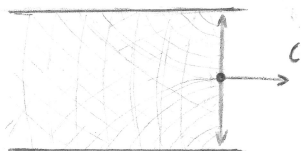
$$\nabla_{\perp}^2 W = q \int_{-\infty}^{\infty} c \frac{d}{dt} \left(\frac{1}{c} \frac{\partial}{\partial t} - \frac{\partial}{\partial z} \right) (c A_s - \phi) dt = q c \cdot \left[\left(\frac{1}{c} \frac{\partial}{\partial t} - \frac{\partial}{\partial z} \right) (c A_s - \phi) \right]_{t=-\infty}^{t=\infty} = 0 \quad \sim \text{since } \vec{A} \text{ and } \phi \text{ vanish at } \pm \infty$$

$$\therefore \nabla_{\perp}^2 W = \frac{\partial^2 W}{\partial x^2} + \frac{\partial^2 W}{\partial y^2} = 0$$

1.5 - The "Causality" Principle

Related to this discussion is the remark that, in the ultra-relativistic limit ($v \rightarrow c$), the wakes generated by a particle can only influence particles that are behind it*, since the EM fields (and thus the "wake wave-front") also propagate at speed c .

Wake analogy:



This is equivalent to saying that W must satisfy:

$$W(\vec{p}_s, \vec{p}_w, z) = 0 \quad \forall \quad z < 0 \quad !$$

Which also implies:

$$\left\{ \begin{array}{l} \bullet \vec{w}_\perp - \vec{\nabla}_{w,\perp} W = 0 \quad \forall \quad z < 0 \\ \bullet w_{||} = \frac{\partial W}{\partial z} = 0 \quad \forall \quad z < 0 \end{array} \right.$$

Even though this is a good approx. for real beams in SL storage rings, the raw effective wakes from time-domain solvers do not respect it, as they are calculated from finite-length charge distributions

The wake formalism also applies to the modelling of other sources of col. effects, such as the interaction of the beam w/ ions in the vacuum chamber, and to model the effect of CSR on the beam ~ former $\rightarrow$ causal $\checkmark$
latter $\rightarrow$ not causal !